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Full Length Article

Friction and wear behavior of chemically modified Sal (*Shorea Robusta*) oil for bio based lubricant application with effect of CuO nanoparticles

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ABSTRACT

Due to degradation of the petroleum reserves in the upcoming future and pollution caused by their usage forces industrialists to search for an environmentally friendly option. Sal oil is having sustainability as a bio based lubricant with nanoparticles as additives. This study investigates the friction and wear characteristics of the chemically modified sal oil with epoxidation process. Furthermore, copper oxide nanoparticles (CuO) were added to the modified oil in different proportions. The lubricants were examined for the thermophysical properties like kinematic viscosity, viscosity index increment, flash point, pour point and dispersion stability with tribological analysis. The test was performed by using a pin on disc tribometer by considering different conditions. Based on the specific concentration (percent) of copper oxide nanoparticles, nanolubricants were properly dispersed through ultrasonication process. During the flow behaviour analysis, all the lubricant samples show Newtonian behaviour by presenting a linear relationship between the shear rate and shear stress. Reduced friction and antiwear properties were obtained at 0.25% and 0.5% concentration of copper oxide nanoparticles The Scanning Electron Microscope (SEM) images also shows better surfaces when the nanoparticle was added up to 0.5% concentration.

1. Introduction

Nowadays, environmental pollution is grooming at a faster rate. Furthermore, energy supplied by petroleum products will also exhausted in the upcoming future due to an increase in energy demand [1–3]. Synthetic lubricants are the petroleum product used for the lubrication of the machine parts.

Friction is generated in the parts during their sliding contact resulting in the generation of heat [4–6]. The life of the parts gets reduced due to the generation of friction. For the reduction of friction, petroleum based lubricants are used. The petroleum based oils are commodities based on hydrocarbons and are included with certain additional ingredients to improve their implementation [7]. However that may be, they have no unfavorable effect on the environment, especially concerning their transfer.

The use of synthetic lubricants is enormous among oil-based commodities due to its application for lubrication reasons [8]. There are some environmental problems associated with the use of hydrocarbon oil since they are accountable for marine pollution owing to spillage after use [9]. There is a need to find an alternative that may be used to replace synthetic oil. Because of this point, the author concentrated on the discovery of a replacement for synthetic oil.

Bio based lubricant is one of the assets and plays a key role in the improvement of sustainability, which is progressively economic and innocuous to nature where it is used. Compared to other sources, choosing vegetable-based lubricant is generally chosen as an alternative in producing the desired lubricants due to its biodegradable properties, higher viscosity, high volatility as well as its lesser toxicity level [10,11]. Improved lubricating properties are mainly attributed to a long-chain of fatty acids, a large amount of unsaturated fatty acid, and polar ester group components in vegetable oils itself. However, various studies were performed while considering vegetable oils for lubricant applications and most of them reported about high wear rate. The presence of unsaturated content in vegetable oil leads to a reduction in oxidation and thermal ability. The main reason is that vegetable-based oils were not properly chemically modified to make it suitable for effective lubrication. The physicochemical properties were important for the use of substitute oils such as viscosity, flash point and fire point in addition to enhancing the physicochemical attributes, nanoparticles are one of the appropriately added substances; numerous tests have been conducted to identify the impact of nanoparticles on tribological attributes, while the previous studies were based on their application to

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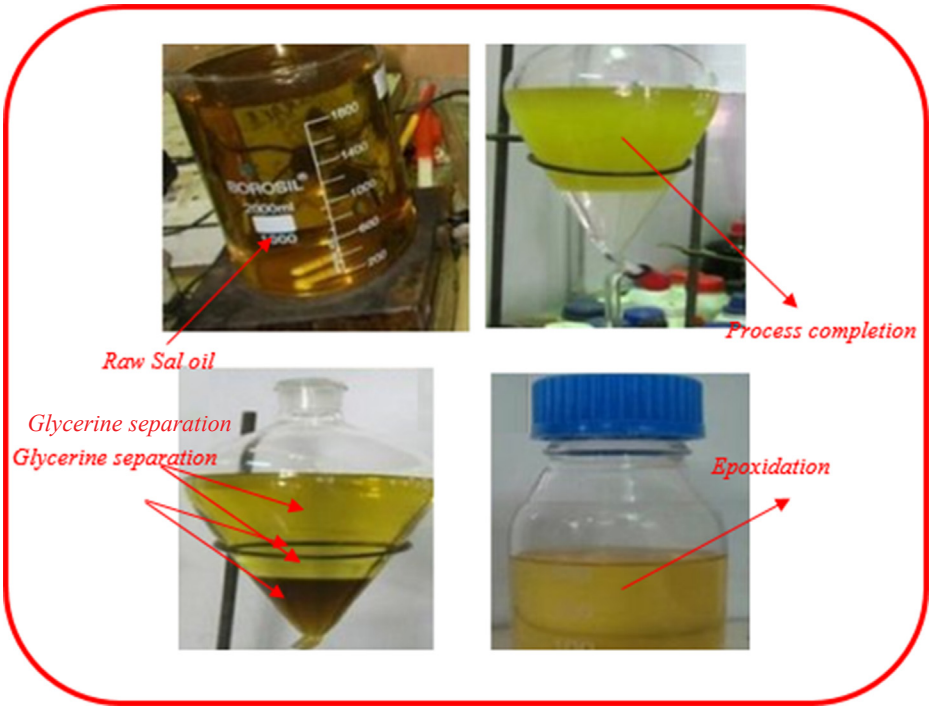


Fig. 1. Image of Sal oil after the epoxidation process.

the conventional petroleum lubricants [12–14]. Xie et al. [15] evaluated the effect of silicon dioxide and molybdenum disulfide nanoparticles on engine oil tribology. Reduction in friction and wear of the parts was observed. Awang et al. [16] evaluated the performance of cellulose nanocrystal to the tribological characteristics of the engine oil. Anti-wear resistance and reduced friction were obtained at the addition of 0.1% concentration. Improvement in the anti-wear mechanism was observed during the application of Aluminium oxide (Al_2O_3) and Silica (SiO_2) nanoparticles to the gear oil. Around 25% and 22% reduction in friction was attained at 0.3% nanoparticles addition [17]. Shi et al. [18] investigated the friction and wear behavior of the graphene/copper hybrid particles. A major amount of reduction in friction was observed during the introduction of hybrid nanoparticles. Borda et al. [19] tested the frictional behaviour of mineral and synthetic oil with the addition of copper. Improved reduction in the quantity of friction coefficient was obtained with mineral oil up to 0.3% addition. Singh et al. [20] performed a tribological analysis by considering copper nanoparticles as an additive to the desert date oil. Better results has occurred at 0.6% concentration. Mamta et al. [21] improved the lubricity characteristics of the neem oil by introducing SiO_2 nanoparticles. Nanoparticles with 0.3% concentration on the oil showed better results. The titanium dioxide (TiO_2) nanoparticles were mixed to the Castor oil for checking its tribological characteristics by Singh et al. [22]. Reduced friction and wear of the surface was occured at 0.3%. It is clear from the previous studies that nano particles are having potential to improve the lubricating properties of the vegetable oils. It has been proved from the literature that the CuO nanoparticles are feasible for the tribological applications and they provided better results through their proper dispersion in the oil. So, the authors have selected CuO based on these conditions. Also, the CuO nanoparticles are having better thermal conductivity which help to reduce the heat from the surface.

In the previous study, none of the research was conducted while considering the application of copper oxide nanoparticles to chemically modified Sal oil. Sal oil is the unattended non-edible oil which has never been used for tribological applications. Sal tree is having a scientific name Shorea Robusta. They are abundantly available in arid to semi-arid regions. They are mostly available in certain countries of Asia such as India, Malaysia, China, and Thailand. Around 1,90,000 tons of oil were produced annually that found its application for paint and pigments [23]. Based on the literature available, the author has decided to conduct a tribological analysis for its possible application in this field. The following are the objectives of this study.

- Chemical modification of Sal oil through epoxidation and the addition of CuO nanoparticles to the modified oil.
- To test the physicochemical properties of the nano lubricants.
- To conduct tribological analysis at the fixed condition for different lubricants.
- To study the worn surfaces through Scanning Electron Microsope assistance.

2. Materials and methods

2.1. Composition of Sal oil

For the evaluation purpose, raw Sal oil was procured from the M/s Pallishree Limited, Kolkata. The oil mainly contains triglyceride esters of fatty acids and glycerol which were estimated according to European standard method EN14103:2003.

The fatty acid composition of Sal oil is analyzed by a Gas Chromotohraphy-Mass Spectroscopy (GC–MS) (GC-10, Shimadzu, Japan) having a capillary column with 30 mm length and 0.25 mm diameter lined with a 0.25 μm (Rtx-5 ms, Rextex). Samples were injected in the split/column flow ratio 24:1. Helium used as the carrier gas (flow rate of 1 ml/min). The injection temperature was 250 °C and column oven 250 °C (programmed to start at 120 °C, held at this temperature for 5 min and heated at a rate of 3 °C/min to 250 °C).

2.2. Chemical modification of the Sal oil

The chemical modification was performed to improve the lubricating properties of the oil. In the chemical modification process, the epoxidation process was performed. To perform this process, acetic acid was mixed with hydrogen peroxide in equal proportions while considering sulphuric acid as a catalyst for obtaining peracetic acid. The proportions selected are based on the work reported by Erween et al.

[24]. Around 10.72 ml of Sal oil was mixed with 37.12 ml of peracetic acid. The mixture was heated at 20 °C for about 4.5 h so that a complete reaction occurs. The obtained solution was emptied into the water which results in the rise of the epoxidized oil to the top surface (Fig. 1). After leaving the mixture for about 1 h, water was extracted from the lower surface of the burette. The remaining oil was heated for about 1 to 2 h to completely remove any moisture remaining. The chemically modified oil gets ready for further testing as bio based lubricant.

2.3. Lubricant properties analysis

The kinematic viscosity was evaluated through the viscometer (M/s Swastik systems and services, New Delhi) based on the ASTM D-445 standard. The viscosity was measured at atmospheric pressure and temperature of 40 °C, 60 °C, 80 °C, and 100 °C. The test was conducted three times and the mean value was used to reduce the error and maintain reliability.

To calculate the viscosity index (VI) of the samples, the following equation (1) was utilized according to the ASTM D-2270:

$$VI = \frac{[L - U]}{[L - H]} * 100 \quad (1)$$

where U represents kinematic viscosity at 40 °C, L and H are the kinematic viscosity at 40 °C with viscosity index 0 and 100, respectively, which have the same kinematic viscosity at 100 °C as the oil whose viscosity index are tried to be ascertained.

The flash point and pour point of the oil were measured according to ASTM D-92 (Cleveland open cup method) and ASTM D-97 respectively, using proper apparatus. The acid value and density were measured based on the standard methods published in the literature [25]. Table 1 shows the properties of the CuO nanoparticles.

The CuO nanoparticles having 40 nm diameter were procured from the M/s Sigma Aldrich, Bangalore. The nanoparticles were mixed to the epoxidized oil on a weight percentage basis (0.25, 0.5 and 1.0). The surface modifier (Triton X-100, M/s Triveni chemical, Gujarat) was used to provide stability to the nanoparticles with the epoxidized oil. The amount of Triton X-100 was 50% (by weight) of each nanoparticle. For attaining steady suspension of the nanoparticles, they are mixed using a magnetic stirrer with 800 rpm/min for 2.0 h duration. Then the mixture was agitated for 40 min through the ultrasonic probe sonicator (M/s Samarth Electronics, Thane). The frequency was 60 kHz and the temperature was set 65 °C for the ultrasonicator. After sonication, the destruction of the molecules occur inside the structure that helps in bonding the nanoparticles to properly dispersed to the lubricant in use.

The dispersion stability of the nanoparticles in the epoxidized oil was analyzed using Zetasizer (Nano ZS, Malvern) procured from ATA Scientific Instruments. The equipment could measure particle size ranging from 0.3 nm to 10 μm.

To examine the size of the nanoparticles, Scanning electron microscope (SEM) having zoom capacity 50× to 300000× was used. Before the examination of the samples by SEM, they are being polished and treated with keller's reagent to provide the clear magnification of the surface. Fig. 2 (a) shows the image of CuO nanoparticles in dry condition and Fig. 2 (b), (c) shows the image of the CuO nanoparticles dispersed in 0.5% and 1.0% respectively. The image clearly shows the amorphous structure and with the help of the metallurgical software equipped within the system, the range was found between 35 and 42 nm. It has been noted that there is no change in the morphology of nanoparticles when dispersed in the lubricant. It has been observed from the Figure that the aggregation size increases with an increase in CuO nanoparticle volume fraction in nano lubricants. This justifies the

Table 1
Properties on CuO nanoparticles.

Materials properties					
Nanoparticle	Purity (%)	Size (nm)	Density (kg/m^3)	Color	Shape
CuO	99.9	40 nm	6341	brown	Spherical

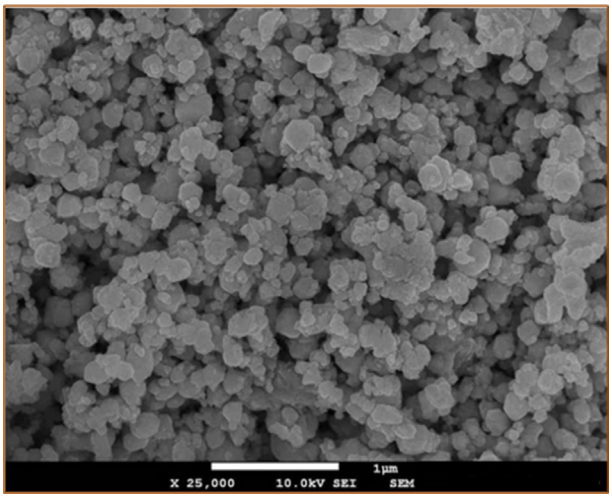


Fig. 2. (a) SEM image of the CuO nanoparticles. **(b).** SEM image of the CuO nanoparticles with 0.5% concentration. **(c).** SEM image of the CuO nanoparticles with 1.0% concentration.

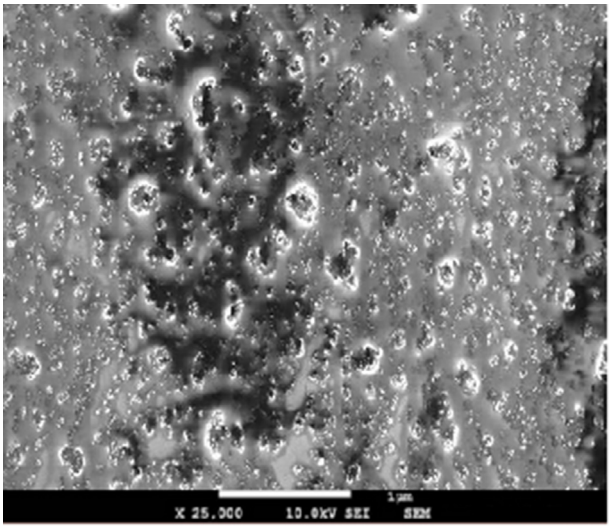


Fig. 2. (*continued*)

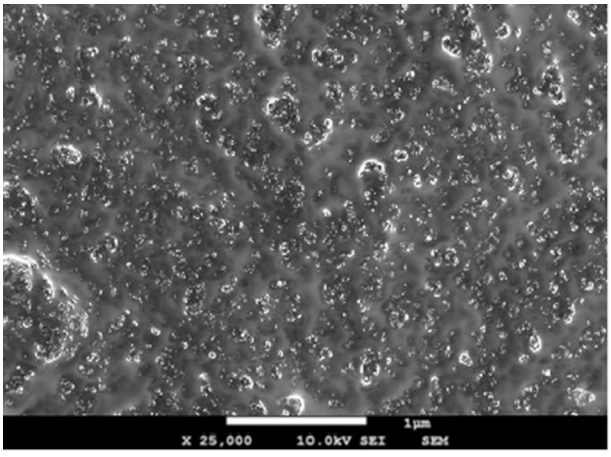


Fig. 2. (*continued*)

results obtained for aggregation by Zetasizer apparatus. Structural properties are investigated by X-Ray Diffraction (XRD) spectroscopy.

Fourier transform infrared (FTIR) spectrometer (M/s Bruker Alpha T) was used to analyze the disappearance of C=C bond and the formation of hydroxyl and epoxy groups in the raw sal oil and the epoxidized sal oil. The wavenumber range was 4000–400 cm^{-1} with a resolution 32 cm^{-1}, 5 mg sample was taken in the crucible having a capacity of 50 mg.

2.4. Tribological analysis

2.4.1. Materials

The LM 16 aluminium alloy (Copper: 1.3%, Manganese: 0.5%, Titanium: 0.2%, Magnesium: 0.5%, Silicon: 5.3%, Iron: 0.6%, Nickel: 0.25%, Zinc: 0.1%, Lead: 0.1%, Tin: 0.05% and remaining Aluminium) was procured from the M/s Samarth Metal Corporation, Maharashtra which was used as pin material for the test. The purpose of consideration of this material due to its application for the piston which faces maximum friction and the hardness is 96 HRB. This type of alloy was also capable to resist wear and corrosion. For the Disc, EN 31 steel was used as it contains more amount of hardness (62 HRC) and it is highly worn resistive. The pin was cylindrically shaped by employing a turning operation on the lathe machine. The one end of the pin was made spherical for attaining point effect with the disc. The pin consists of 8 mm diameter and length was 30 mm. The pin was further polished by using emery paper of following grit sizes 200, 400 600, and 1200 nm before experimenting on the machine.

2.4.2. Experimental set up

The pin on disc machine was purchased from the M/s DUCOM, Bangalore, India that was used for investigating the friction and wear characteristics during the application of lubricants. Fig. 3 shows the schematic image of the mechanism involving point contact of pin and disc. The friction force was obtained on the screen with the help of the load cell attached to the tribo machine. For obtaining the friction coefficient, friction force was normalized by the load applied. The lubricant was provided in a drop wise manner to the interfaces with the help of the pump operated by the electric motor. Table 2 mentioned the conditions considered for the examination of the samples. Table 2 showed the load and sliding speed taken for the test. The conditions were considered based on previous studies [26,27]. The maximum load on the engine during its working comes under idle condition or when the speed is minimum. So, the author didn't consider more load and speed.

Table 2
Condition taken during the analysis.

Conditions considered	Value
Applied load (N)	110
Track diameter (mm)	80
Velocity (m/s)	4.2
Temperature (°C)	75
Duration (s)	3600

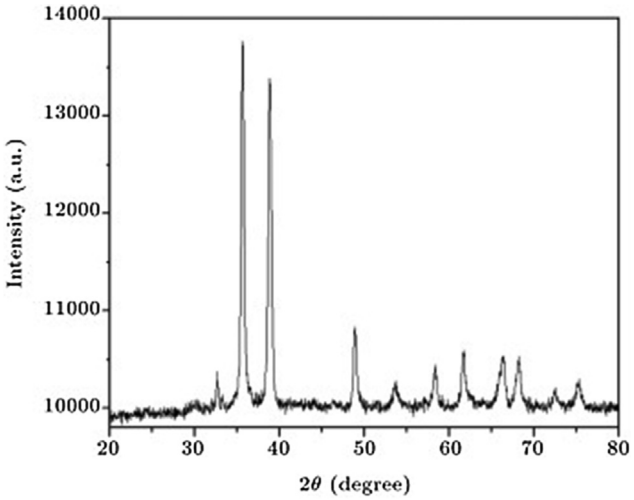


Fig. 4. XRD pattern of CuO nano powder.

3. Results and discussion

3.1. Structural analysis of CuO nano powder

Fig. 4 shows the XRD image of the CuO nano powder, According to

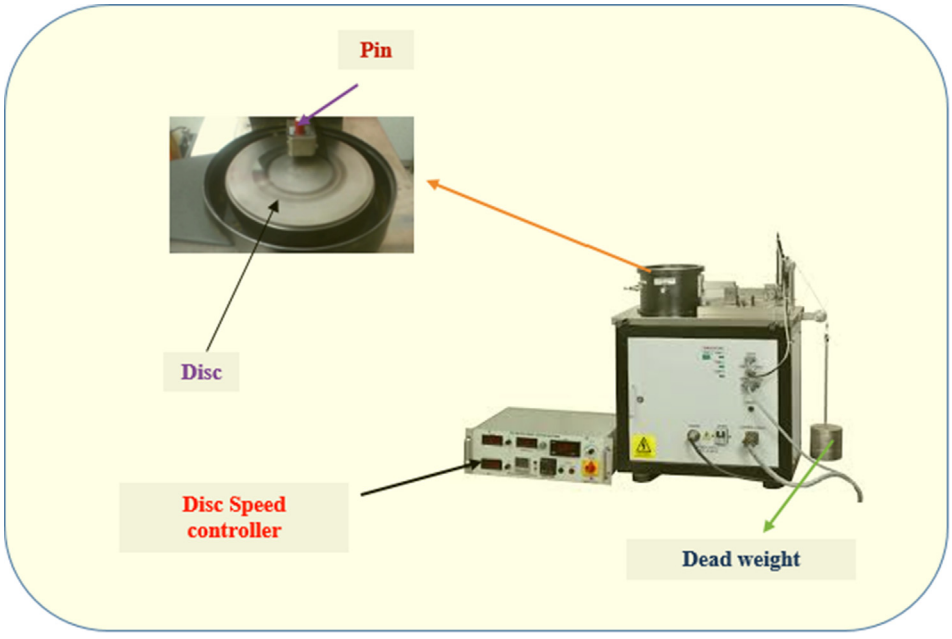


Fig. 3. Image of the set up for the tribological analysis.

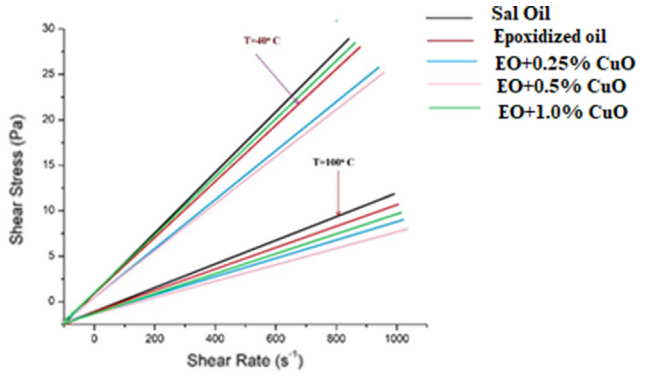


Fig. 5. Shear rate with variation of shear stress for different lubricants.

the literature [28], two reflections at 2θ = 34.8 [0 0 2] and 2θ = 37.6 [1 1 1] were observed in the diffraction patterns, and are ascribed to the formation of the CuO (space group C2/c) monoclinic crystal phase.

3.2. Rheological analysis

Fig. 5 shows the shear stress variation with the shear rate for all the lubricant samples. For both the temperature, the trend comes out linear which confirms the Newtonian characteristics of the samples considered for the study. With an increase in temperature, the shear rate increases. The maximum shear rate occurred at a higher temperature 100 °C. The Sal oil shows the maximum shear rate concerning expoxidized oil. The further reduction in the shear rate value was obtained with the addition of the nanoparticles to the epoxidized oil. With an increase in nanoparticle concentration after 0.5%, a more shear rate was observed. This was due to the breakdown of the lubricant layers associated with the fluid. The change in viscosity of the samples concerning the shear rate is depicted in Fig. 6. There are negligible changes observed in the viscosity of the tested samples that confirms their Newtonian behaviour. The viscosity of the lubricant decreases with an increase in temperature. The higher viscosity was found at epoxidized oil which further increases with the concentration of nanoparticles. The nanoparticles mixed with fluid at a molecular level that results in higher viscosity [16,29]. The same trend has also been observed in the study done by Kerni et al. [30].

3.3. Fatty acid composition analysis

The most dominant factor in the raw Sal oil is the presence of Stearic acid contributing to around 47.21%. Around 38.19% amount of Oleic acid is also present in the raw oil. Fig. 7 shows the number of fatty acids present in the Sal oil. [31]. Palmitic acid, Arachidic acid, and Linoleic acid composition is 2.17%, 3.82% and 1.72% respectively. The presence of a major amount of stearic acid depicts the presence of unsaturated fatty acids that further needs chemical modification for lubrication application. The amount of fatty acid was calculated based on the following equation. The fatty acid for the raw Sal oil was 6.9%. The fatty acid comes above the standard for the lubrication needed so, the chemical modification was performed [32,33].

$$FFA = \frac{Acidnumber}{2} \quad (2)$$

where, FFA = Free fatty acid.

Fig. 6. Viscosity for shear rate for different lubricant samples.

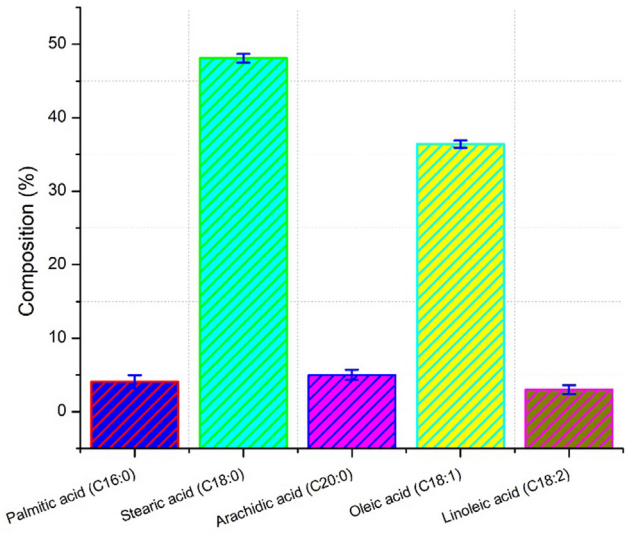


Fig. 7. Fatty acid composition of Sal oil.

3.4. FTIR analysis

Fig. 8 shows the FTIR spectrum evaluated for the raw oil and epoxidized Sal oil. The presence of fatty acids with double bonds can be observed in the range of 2800–3000 cm^{-1}. The C=C peak was observed at 1653 cm^{-1} wavenumber which got disappeared in the epoxidized FTIR spectrum indicating its consumption. Due to the epoxidation process, evidence of the bond C-O-C was observed at 823 cm^{-1}. Similar kinds of results were also reported in the FTIR spectrum of epoxidized olive oil in the literature [30]. The presence of hydroxyl group (–OH-) in the epoxidized oil can be observed at wavenumber 3464 cm^{-1}.

3.5. Physicochemical properties analysis

Table 3 shows the physicochemical properties of the lubricants. The kinematic viscosity of raw sal oil was 36.3 Cst at a temperature of 40 °C and 11.2 Cst at 100 °C. There was an increase in kinematic viscosity obtained with modification of the raw oil. This is due to the dissolution of the double bonds associated with raw vegetable oil [34].

The results of the viscosity index (VI) indicates the temperature range that lubricant can resist during the operation. A high VI value means there is a little change of kinematic viscosity along with the change of temperature. Meanwhile, the higher value of VI ensures that lubricant samples provide good and stable lubricity through a wide range of operating temperatures. According to Zulkifli et al. [35], the lubrication film becomes extremely thin at high temperatures and very thick at low temperatures. The modification of raw sal oil improved the value of VI.

Fig. 9 shows the size distribution of dispersed nanoparticles in nanolubricant samples using Zetasizer (Nano ZS, Malvern). It has been observed that the average size of CuO nanoparticles based on the concentrations 0.25%, 0.5% and 1.0% are 147.1, 156.1 and 204.1 nm respectively, which is approximately 3 times higher than the original size. This increase in size after the dispersion of nanoparticles is due to the mild agglomeration of nanoparticles. The size of CuO nanoparticles is higher after mixing due to the amorphous morphology of CuO nanoparticles [36].

3.6. Effect of nanoparticles

3.6.1. Viscosity increment

Fig. 10 shows the viscosity increment in percentage when the CuO nanoparticles are added to the epoxidized oil in different proportions. It

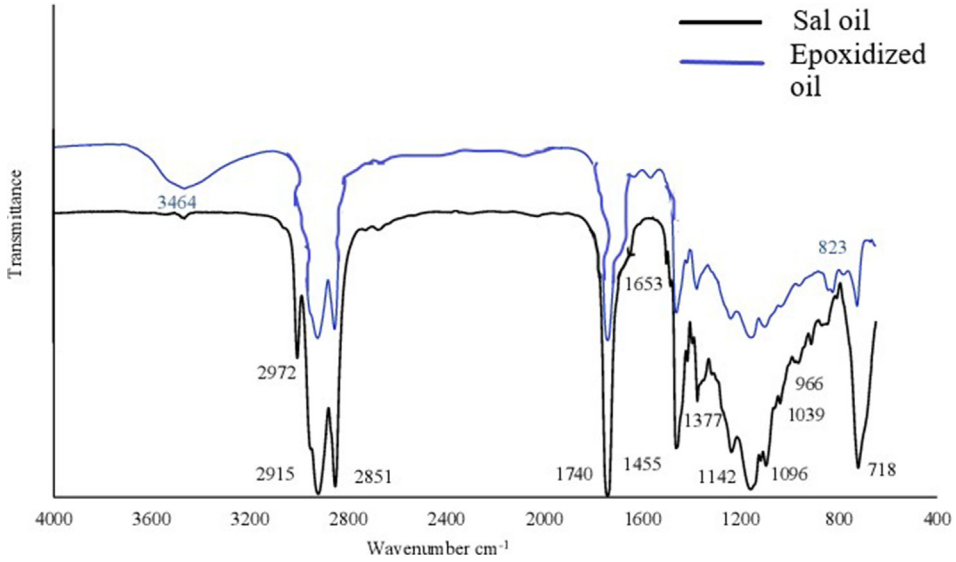


Fig. 8. FTIR spectrum a) raw sal oil and (b) epoxidized sal oil.

Table 3
Properties of the Sal oil and epoxidized oil.

S. No.	Properties	Sal oil	Epoxy Sal oil
1.	Viscosity (Cst) @ 40 °C	38.3	104.2
2.	Viscosity (Cst) @ 100 °C	10.29	11.72
3.	Viscosity index	118	203
4.	Density (kg/m^3) @ 15 °C	914.8	902.6
5.	Flash point (°C)	123.8	169.2
6.	Pour point (°C)	−2.9	1.7
7.	Acid value	12.8	0.078

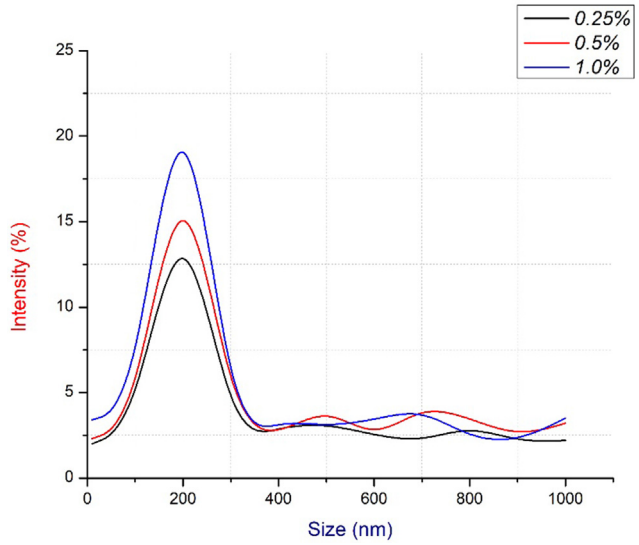


Fig. 9. Size distribution of dispersed CuO nanoparticles to expoxidized oil.

can be seen that the addition of the nanoparticles increases the viscosity with comparison to the epoxidized oil. The viscosity of all the lubricants decreases with an increase in temperature. The maximum increment in viscosity was found at 1.0% and 100 °C. The increase of nanoparticles results in agglomeration on the surface and also molecular collision occurs which increases nano lubricant viscosity. The collision of the molecular forces with an increased temperature reduces the viscosity of the nano lubricant [37]. The mathematical relation for the viscosity increment measurement is calculated by using Equation (3).

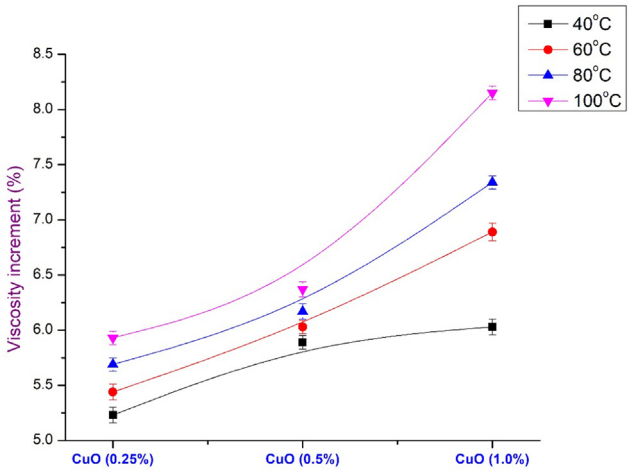


Fig. 10. Viscosity increment for different CuO nanoparticles concentration.

$$Viscosity\ increment\,(\%) = \frac{Nano\ lubricant\ viscosity - Epoxidized\ oil\ viscosity}{Epoxidized\ oil\ viscosity} * 100 \quad (3)$$

3.6.2. Relative viscosity

The relative viscosity with change in temperature for different concentrations is shown in Fig. 11. The maximum change can be observed for the 1.0% copper nanoparticles addition at maximum temperature 100 °C. Eq. (4) shows the mathematical relation for getting relative viscosity. From the relation, it can be observed that the increment in relative viscosity is dependent on the fluid viscosity considered for the analysis. As the viscosity of the fluid increases with the addition of the nanoparticles so, a higher amount of rising is observed in relative viscosity with 1.0% CuO nanoparticles addition [38].

$$\mu_r = \frac{\mu_{nl}}{\mu_{eo}} \quad (4)$$

where, μ_r = relative viscosity; μ_{nl} = viscosity of nano lubricant; and μ_{eo} = viscosity of epoxidized oil.

3.6.3. Viscosity index increment

Viscosity index is another important parameter that decides the lubricity and it is based on the kinematic viscosity of the samples.

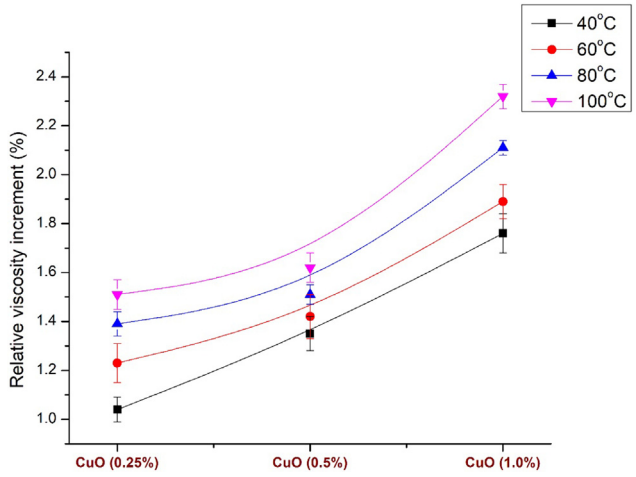


Fig. 11. Variation of relative viscosity with temperature.

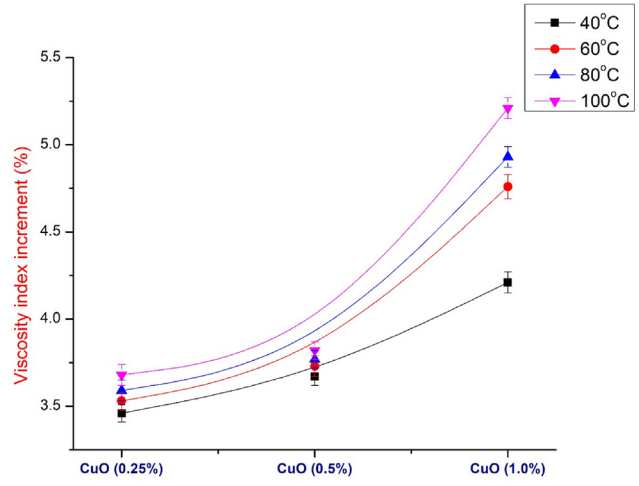


Fig. 12. Viscosity index increment concerning the CuO nanoparticles at various temperatures.

Fig. 12 shows the variation of viscosity index increment concerning the temperature for various concentrations of the nanoparticles. The maximum viscosity during the addition of the nanoparticles was obtained at 1.0% and temperature 100 °C. The minimum changes in the viscosity index increment were observed at lower concentrations and temperatures. These changes occurred due to higher molecular collision occurrence at higher temperature [39].

3.6.4. Flash and pour point

Fig. 13 shows the effect of CuO nanoparticles on the (a) flash point and (b) pour point. With an increase in the concentration of the nanoparticles, an increase in the value of the flash point was observed which was a good indication. The increase in flash point provides an increased operating limit of the temperature of the lubricants. Due to an increase in the limit of operating temperature, more stability and better thermophysical properties of CuO nanoparticles were observed. Hence, the enhanced flash point can be considered as an advantage about enhanced lubricity characteristics of epoxidized oil. With an addition of the nanoparticles, pour point increases up to 0.5% concentration and its reduction was obtained at 1.0% concentration. The increase in temperature resists to the flow of the lubricant and a higher concentration of the nanoparticles doesn't provide a better flow [8,16,40,41].

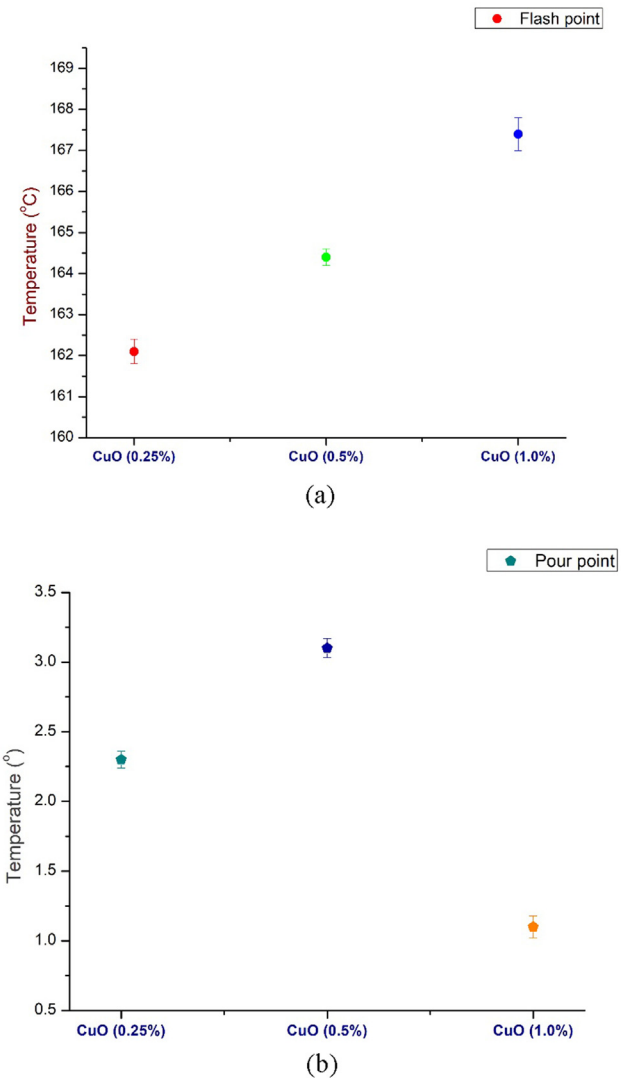


Fig. 13. Effect of nanoparticles addition on the (a) flash point and (b) pour point of the nano lubricant.

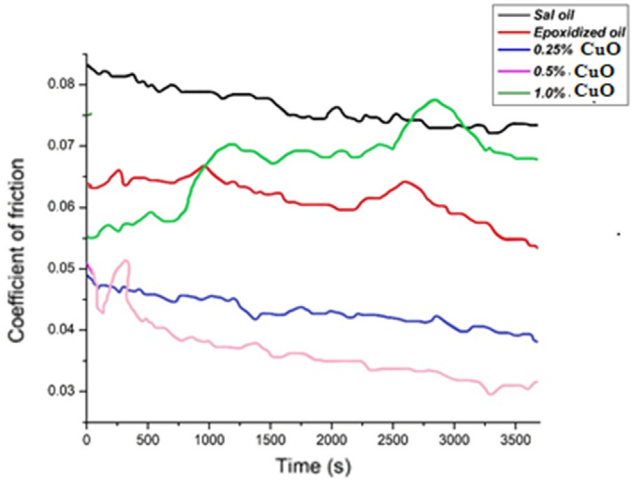


Fig. 14. Coefficient of friction of nanolubricant samples.

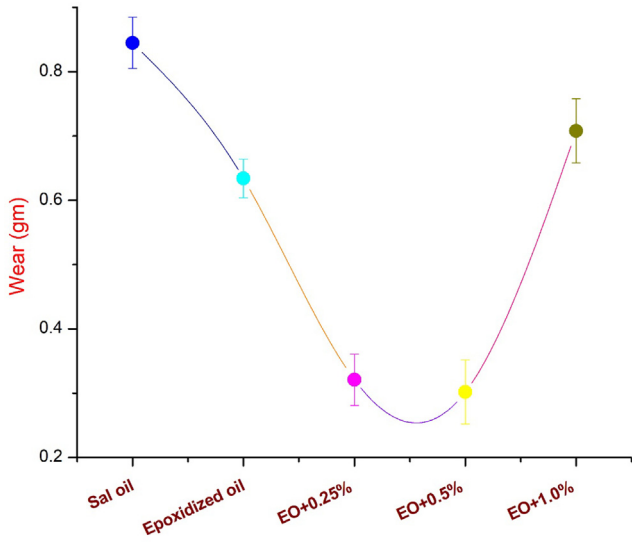


Fig. 15. Wear for different samples.

3.7. Frictional analysis

Fig. 14 shows the frictional behavior of the lubricants. It has been observed that epoxidized Sal oil presented a minimum friction coefficient for the raw oil. This was due to the formation of a thick film that was created by the tribochemical reaction. It has been reported that the formation of the oxirane ring provides an effective layer on the surfaces during the motion. The raw oil initially shows a minimum coefficient of friction which showed increased behavior after 500 s due to the instability of the film formed on the surfaces.

The effect of nanoparticles addition is having better results. The minimum coefficient of friction was obtained when a concentration of 0.5% nanoparticles is there in the chemically modified oil. The coefficient of friction was reduced by about 8% when compared to the raw oil. The mechanism of friction reduction involves the filling of gaps on the surfaces with nanoparticles which provides smoother and better lubricity. With the addition of the nanoparticles, contact surface area increased which resulted in a reduction of the applied pressure and rolling mechanism occurs instead of sliding friction. The same reason has also been reported by Ghaednia et al. [28]. With an increase in the amount of nanoparticles after a limit, particles get aggregated on the metals which enhances friction and wear on the surface during their contact. The same hypothesis has also been reported in the previous studies and also presenting abrasive wear with increased nanoparticle concentration [27,41,42].

3.8. Wear analysis

The wear of the pin was minimum for the epoxidized Sal oil with comparison to the base oil as evident from Fig. 15. The epoxidized oil forms an improved lubricant layer between the surfaces during their contact. The epoxidized oil contains poly esters which promote

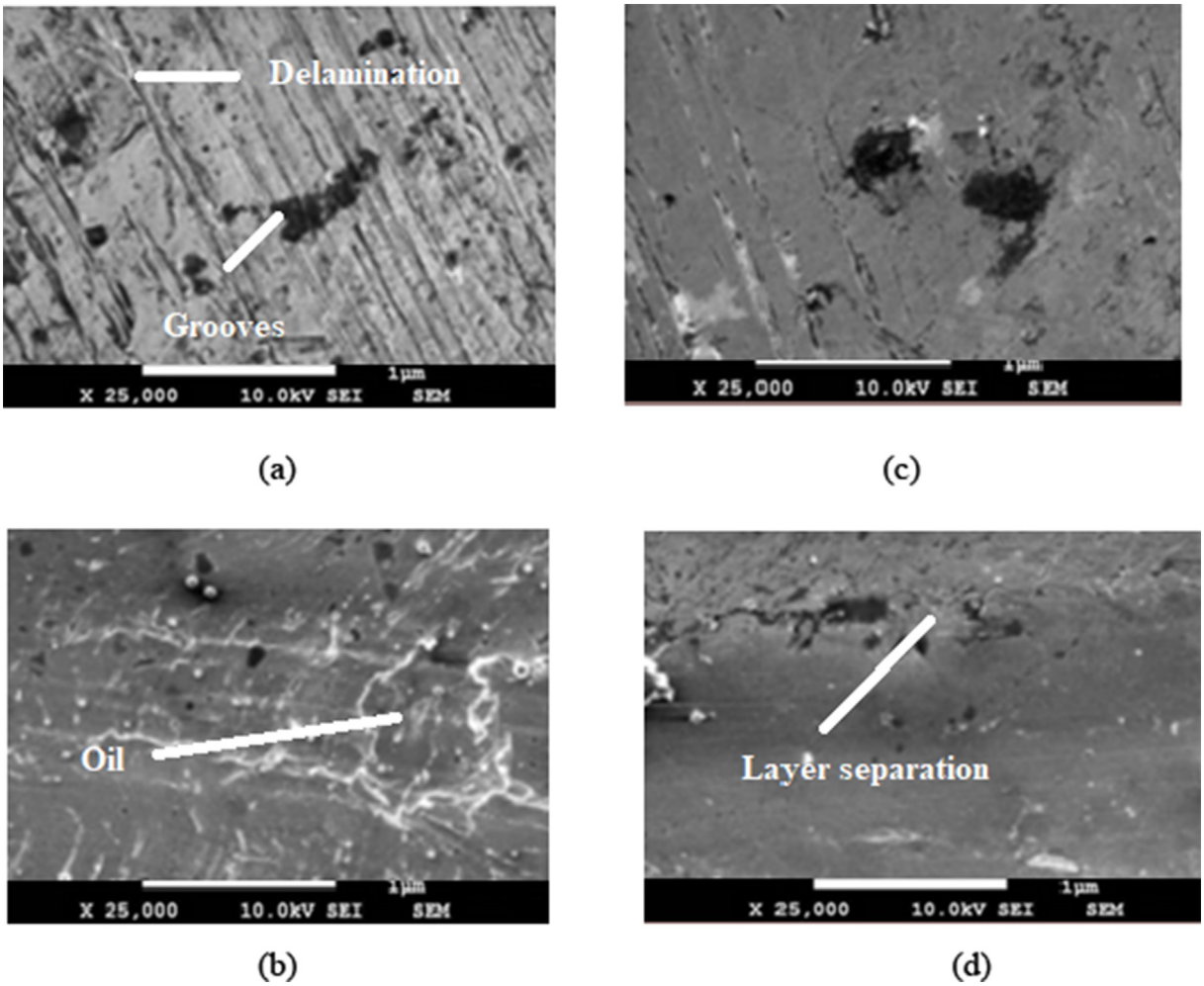


Fig. 16. SEM images of the lubricants tested (a) Raw Sal oil (b) Epoxidized Sal oil (c) EO + 0.5% CuO (d) EO + 1.0% CuO.

adsorption on the surface resulting in minimum wear [20].

The concentration of CuO nanoparticles up to 0.5% also showed minimum wear of the pin when compared to the raw Sal oil. The reduction in wear was around 4.2%. The nanoparticles effect on the anti-wear property depends on the proper dispersion of the particles in the lubricant. The CuO nanoparticles are having the capability to gets properly dispersed in the solution which promotes anti-wear mechanism. These results are also justified by the SEM images of the samples considered for the analysis.

3.9. Surface topography

Fig. 16 shows the SEM images of the lubricant samples obtained after performing the test. Fig. 16 (a) shows the SEM images for the Sal oil. The surface layer got delaminated due to the failure of the lubricant layer. This was due to the stress developed on the surface that results in the deformation of the layers. Fig. 16 (b) shows the surface at Epoxidized oil. The surface observed was smoother than the raw Sal oil due to the formation of a better lubricant layer. This was due to the formation of -O- cross-linking on the surface which protects it by assisting the lubricant film [43–45]. The addition of the nanoparticles up to 0.5% results in less wear as they provide a defensive film between the surfaces during their motion. Fig. 16 (c) shows SEM images with the effect of 0.5% copper oxide nanoparticles addition. The minimum damage to the surface was observed which was due to improvement in the layer of the epoxidized oil with the addition of the nanoparticles. The nanoparticles mixed with the lubricant at a molecular level which assists the layer to resist the stress [46]. However, a further increase in nanoparticles results in more amount of wear. Fig. 16 (d) shows the effect of adhesion and abrasion on the surface when the amount of copper oxide nanoparticles further increased to 1.0%. The nanoparticles when increases to after a certain limit, they get agglomerated on the surface. The flow of the lubricant from that region also provides pressure to the nanoparticles resulting in a more abrasive nature [16].

4. Conclusions

- From the rheology test, it has been confirmed that the nano lubricants follow Newtonian behavior.
- The improved viscosity was found with epoxidized oil, epoxidized oil with 0.5% concentration.
- With the addition of the nanoparticles at 0.5%, an increase in the viscosity index percentage was observed. A higher amount of increment was obtained at higher concentrations of 1.0%.
- The addition of the nanoparticles to the chemically modified oil increases the flash point of the nano lubricants. A higher value of the flash point was obtained at 1.0% which was around 3.42% increment from the epoxy oil.
- The addition of the CuO nanoparticles to the chemically modified oil improves the lubricity of the samples. With the addition of up to 0.5% concentration, better results are obtained in terms of reducing friction.
- The addition of the nanoparticles up to 0.5% to the modified oil shows less wear and damage to the surface due to the formation of a better constructive film on the surface.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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